

Engineering Probability

Homework 2

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(1)

A machine produces photo detectors in pairs. Tests show that the first photo detector is acceptable with probability $3/5$. When the first photo detector is acceptable, the second photo detector is acceptable with probability $4/5$. Otherwise, if the first photo detector is defective, the second photo detector is acceptable with probability $2/5$.

- (a) What is the probability that exactly one photo detector of a pair is acceptable?

A = detector 1, B = detector 2.

$$P(\text{singular detector}) = P(A \cap B^c) + P(A^c \cap B)$$

$$P(A \cap B^c) = P(B^c|A) \times P(A) = \frac{1}{5} \times \frac{3}{5} = \frac{3}{25} = 0.12$$

$$P(A^c \cap B) = P(B|A^c) \times P(A^c) = \frac{2}{5} \times \frac{2}{5} = \frac{4}{25} = 0.16$$

$$P(\text{singular detector}) = 0.12 + 0.16 = 0.28$$

- (b) What is the probability that both photo detectors in a pair are defective?

$$P(A^c \cap B^c) = P(B^c|A^c) \times P(A^c) = \frac{3}{5} \times \frac{2}{5} = \frac{6}{25} = 0.24$$

(2)

Box 1 contains 1 white and 999 red balls. Box 2 contains 1 red and 999 white balls. A ball is picked from a randomly selected box. If the ball is red, what is the probability that it came from box 1?

$$P(\text{Box}_1|\text{Red}) = \frac{P(\text{Box}_1 \cap \text{Red})}{P(\text{Red})} = \frac{\frac{999}{2000}}{\frac{1000}{2000}} = 0.999$$

(3)

Box 1 contains 1000 bulbs of which 10 percent are defective. Box 2 contains 2000 bulbs of which 5 percent are defective. Two bulbs are picked from a randomly selected box

- (a) Find the probability that both bulbs are defective.

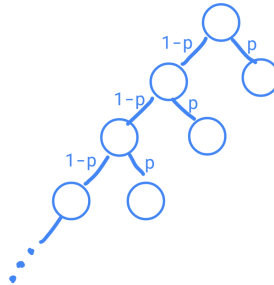
$$P(\text{Both defective}) = \frac{1}{2} \left(\frac{100}{1000} \times \frac{99}{999} \right) + \frac{1}{2} \left(\frac{100}{2000} \times \frac{99}{1999} \right) \approx 0.00619$$

- (b) Assuming that both are defective, find the probability that they came from Box 1.

$$P(\text{Both Box}_1 \mid \text{Both defective}) = \frac{P(\text{Both Box}_1 \cap \text{Both defective})}{P(\text{Both defective})} = \frac{\frac{1}{2} \left(\frac{100}{1000} \times \frac{99}{999} \right)}{0.00619} \approx 0.800$$

(4)

Each time a fisherman casts his line, a fish is caught with probability p , independent of whether a fish is caught on any other cast of a line. The fisherman will fish all day until a fish is caught and then he quits and goes home. Let C_i denote the event that on cast i the fisherman catches a fish. Draw the tree for this experiment and find the following probabilities:



- (a) $P(C_1) = p$
(b) $P(C_2) = (1 - p) \times p$
(c) $P(C_n) = (1 - p)^{n-1} \times p$

(5)

There is a collection of field goal kickers, which can be divided into two groups 1 and 2. Group i has $3i$ shooters. On any kick, a kicker from group i will kick a field goal with probability $1/(i + 1)$, independent of the outcome of any other kicks by that kicker or any other kicker.

- (a) A kicker is selected at random from among all the kickers and attempts one field goal. Let K be the event that a field goal is kicked. Find $P(K)$.

$$P(K) = P(\text{group 1} \cap K) + P(\text{group 2} \cap K) = \left(\frac{3}{9} \times \frac{1}{1+1}\right) + \left(\frac{6}{9} \times \frac{1}{2+1}\right) = \frac{7}{18} = 0.38\bar{8}$$

- (b) Two kickers are selected at random. For $j = 1, 2$, let K_j be the event that kicker j kicks a field goal. Find $P(K_1 \cap K_2)$. Are K_1 and K_2 independent events?

$$P(K_1 \cap K_2) = P(K_1 \cap K_2 | G_1 G_1)P(G_1 G_1) + P(K_1 \cap K_2 | G_2 G_2)P(G_2 G_2) + P(K_1 \cap K_2 | G_1 G_2)P(G_1 G_2)$$

$$\left(\frac{1}{2}\right)^2 \left(\frac{3}{9} \times \frac{2}{8}\right) + \left(\frac{1}{3}\right)^2 \left(\frac{6}{9} \times \frac{5}{8}\right) + \left(\frac{1}{2} \times \frac{1}{3}\right) \left(2 \times \frac{6}{9} \times \frac{3}{8}\right) = \frac{65}{432} \approx 0.150$$

$$\left(\frac{7}{18}\right)^2 = \frac{49}{324} \neq \frac{65}{432}, \therefore K_1 \text{ and } K_2 \text{ are not independent}$$

- (c) A kicker is selected at random and attempts 10 field goals. Let M be the number of misses. Find $P(M = 5)$.

$$P(M = 5) = P(M = 5 | G_1)P(G_1) + P(M = 5 | G_2)P(G_2)$$

$$P(M = 5 | G_1) = \binom{10}{5} \times \left(\frac{1}{2}\right)^5 \left(1 - \frac{1}{2}\right)^5 = \frac{63}{256} \approx 0.246$$

$$P(M = 5 | G_2) = \binom{10}{5} \times \left(\frac{1}{3}\right)^5 \left(1 - \frac{1}{3}\right)^5 = \frac{8064}{59049} \approx 0.137$$

$$P(M = 5) = \left(\frac{63}{256}\right)\left(\frac{1}{2}\right) + \left(\frac{8064}{59049}\right)\left(\frac{1}{3}\right) \approx 0.191$$